# 1.4 Trigonometric Functions of Any Angle

Name _____ Period ____ Date _____

Pre-Calculus 1.4 · Student sheet

**I can…**

- evaluate trigonometric functions from a point on the terminal side.
- find the sign of a trigonometric function from the quadrant.
- find reference angles and use them to find exact values.

**Vocabulary** · fill in the missing word

**Reference angle** — the acute angle $\theta'$ between the _____ side of $\theta$ and the x-axis

**Quadrantal angle** — an angle whose _____ side lies on an axis

## Part 1 · Signs by quadrant

$r = 5$ for each point.

| Point | Quadrant | $\sin\theta$ | $\cos\theta$ | $\tan\theta$ |
|---|---|---|---|---|
| $(3, 4)$ | | | | |
| $(-3, 4)$ | | | | |
| $(-3, -4)$ | | | | |
| $(3, -4)$ | | | | |

## Part 2 · Notes and examples (fill in each note, then work the example)

**Any angle:** for $(x, y)$ on the terminal side, $r = \sqrt{(x^2 + y^2)}$: $\sin\theta = y/r$, $\cos\theta =$ _____, $\tan\theta = y/x$.

### Example 1

$(-3, 4)$ is on the terminal side of $\theta$. Find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

### Example 2

$\tan\theta = -\frac{5}{4}$ and $\cos\theta > 0$. Find $\sin\theta$ and $\sec\theta$.

### Example 3

Evaluate $\cos\pi$, $\tan 0$, and $\sin\frac{3\pi}{2}$.

**Reference angle $\theta'$:** the acute angle between the terminal side and the _____.

### Example 4

Find the reference angle.

a. $300°$

b. $2.3$

c. $-135°$

**To evaluate:** find the value for $\theta'$, then give it the _____ of the quadrant.

### Example 5

Evaluate.

a. $\cos \frac{4\pi}{3}$

b. $\tan(-210°)$

c. $\csc \frac{11\pi}{4}$

## Part 3 · Partner practice

### Partner 1

$(-2, -3)$ is on the terminal side of $\theta$. Find $\sin \theta$, $\cos \theta$, and $\tan \theta$.

### Partner 2

Evaluate $\sin 210°$.